



CHAPTER

About this manual

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1.1 Purpose and scope

This manual describes the SalinBloc™ container: a modular SonixED™ electro dialysis unit for water desalination and brine valorization. The standard product is a single container fitted with the stack, dosing and control equipment; an optional second container adds pretreatment ahead of it for sites whose raw water needs filtering first (chapters 3 and 8). Part I of this manual covers the unit as a whole (safety, installation, commissioning, day-to-day operation). Part II gives one self-contained chapter per compartment inside the container — pumps, stacks, valves, dosing, sensors, control cabinet, the diluate, concentrate, CIP and electrolyte tanks (each its own chapter, since they hold very different streams — and, for the CIP tank, very different chemicals at different times — under very different conditions), and telemetry — each with its own specifications, operating notes, maintenance and troubleshooting, so it can be consulted (or printed/exported) on its own. Part III closes with shutdown/storage and reference appendices.

SonixED™ is HydroVolta NV's electro dialysis-reversal (EDR) technology, combined with an integrated ultrasound antifouling stage and a custom ion-selective membrane coating. HydroVolta NV (Leuven, Belgium, founded 2016) developed SonixED™ over eight years of lab validation with research partners including VITO, and holds a granted European patent (2021) on the combined EDR + ultrasound process. SonixED™/SalinBloc™ units are in active use or pilot deployment at sites in Belgium, Morocco, Australia (Perth), Brazil and Saudi Arabia; see section 3.2 for field performance data published by HydroVolta.



NOTICE

Several sections in this manual are marked **TODO**. These mark information that was not found in HydroVolta's engineering documentation or public materials at the time of writing (July–August 2026) and must be supplied by HydroVolta engineering, or confirmed by inspection/measurement on an actual unit, before this manual is issued to a customer. Every other statement in this manual is grounded in a specific HydroVolta engineering document, datasheet, or the hydrovolta.com website — but the underlying documents mostly describe pilot/demonstration containers (Farys, Ganzepoot, BASF, Morocco, and others), not a finalized, branded “SalinBloc™” bill of materials. Treat pilot-specific facts as a well-informed engineering basis for the standard product, not as a guaranteed as-shipped specification, until confirmed against a specific unit's documentation. Notably, the **PLC control program itself is out of scope for this edition** of the manual — it is still under development by HydroVolta's automation team, and the relevant sections (HMI screen navigation, exact control logic, full setpoint list) will be completed once that work is finished.

1.2 Who this manual is for

This manual is written for the people who take delivery of, install, commission, operate and maintain a SalinBloc™ unit: site engineers, installation contractors, plant operators, and HydroVolta service technicians. Readers are assumed to have basic familiarity with industrial piping, low-voltage electrical systems and PLC-controlled equipment; no prior knowledge of electrodialysis is assumed (see the Glossary, appendix D, for unfamiliar terms).

1.3 Symbols and signal words

Hazard and information boxes in this manual follow the signal-word convention below, aligned with ISO 3864 / ANSI Z535 safety-color practice.



DANGER

Indicates a hazard that, if not avoided, **will** result in death or serious injury.



WARNING

Indicates a hazard that, if not avoided, **could** result in death or serious injury.



CAUTION

Indicates a hazard that, if not avoided, could result in minor or moderate injury.



NOTICE

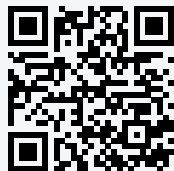
Indicates a risk of property damage or equipment malfunction, with no personal-injury hazard.

NOTE

Highlights additional information that helps the reader but is not itself a hazard warning.

1.4 Online documentation and QR code

A QR code on the container's control cabinet links to the latest online version of this manual and to the unit's factory quality report.



Scan to open the online manual and this unit's quality report at <https://hydrovolta.com/salinbloc-manual>.

1.5 Contact and support

Hydrovolta NV, Romeinse Straat 18, 3001 Leuven, Belgium. E-mail: info@hydrovolta.com. Phone: +32 28 800 336.
Web: hydrovolta.com. See section 7.6 for when to contact service.